

MILESTONE 2: NETWORK EXPANSION

CISCO ENTERPRISE ARCHITECTURE & SECURITY DEPLOYMENT

Project Information

Subject: ICS 605 – Network Security
Faculty: Informatics and Computer Science
Major: IT Security
Instructor: Dr. Marwa Zamzam
TA: Ms. Salma Abo Bakr
Date: May 13, 2026

Implementation Team

Name	Student ID
Eyad Ahmed	13005238
Abdelomen sayed	13006494
Ebram hany	13002244
Yassin azab	13002279
Ali Abdallah	13006577

Contents

1	Project Overview & Scope	2
1.1	Subnet Allocation	2
2	Network Architecture Planning	2
2.1	New Branch IP Addressing Table	2
2.2	Physical Connection Matrix	3
3	Phase 1: Physical Infrastructure Staging	4
4	Phase 2: Core & Switching Configuration	6
5	Phase 2: Core & Switching Configuration	8
6	Phase 3: Centralized Security Services	10
7	Phase 4: Routing & Connectivity Verification	11
8	Phase 5: Advanced Security Policies	13
9	Configuration Verification & Evidence	15
9.1	Configuration 1: OSPF Routing Protocol	15
9.2	Configuration 2: Site-to-Site VPN	20
9.3	Configuration 3: AAA TACACS+ Server Authentication	23
9.4	Configuration 4: Centralized SYSLOG Logging	25
9.5	Configuration 5: Router Brute-Force Prevention	26
9.6	Configuration 6: Switch Port Security Rules	27
9.7	Configuration 7: NVRAM Configuration Archival	32
9.8	Configuration 8: Secure Enable Secret Implementation	33
9.9	Configuration 9: Universal Hashing of Passwords	34
9.10	Configuration 10: Subnet Allocation (192.168.5.0/24)	35
10	Conclusion	37

1 Project Overview & Scope

This document details the expansion of our enterprise network infrastructure to include a fully operational branch office. Milestone 2 focuses on high-availability switching, inter-branch connectivity via Serial WAN, and the implementation of advanced security protocols including Site-to-Site VPN, AAA (TACACS+), and logging.

1.1 Subnet Allocation

The new branch operates on a unified 192.168.5.0/24 subnet for internal operations, while utilizing a specific segment of the 172.16.3.0/24 range for the ISP serial link to avoid overlap with the existing DMZ.

2 Network Architecture Planning

2.1 New Branch IP Addressing Table

Device	Label	Interface	IP Address	Mask	Gateway
Edge Router	R4	Se0/0/0	172.16.3.254	255.255.255.0	—
Edge Router	R4	Gi0/0	192.168.5.1	255.255.255.0	—
ISP Router	R3	Se0/0/1	172.16.3.253	255.255.255.0	—
Core Switch	CS	Vlan 1	192.168.5.2	255.255.255.0	192.168.5.1
IT Switch	B-IT-SW	Vlan 1	192.168.5.5	255.255.255.0	192.168.5.1
Emp Switch	B-Emp-SW	Vlan 1	192.168.5.6	255.255.255.0	192.168.5.1
AAA Server	B2-AAA	Fa0	192.168.5.3	255.255.255.0	192.168.5.1
Syslog Server	B2-Log	Fa0	192.168.5.4	255.255.255.0	192.168.5.1
IT PC 1	B-IT-PC1	Fa0	192.168.5.10	255.255.255.0	192.168.5.1
IT PC 2	B-IT-PC2	Fa0	192.168.5.11	255.255.255.0	192.168.5.1
IT PC 3	B-IT-PC3	Fa0	192.168.5.12	255.255.255.0	192.168.5.1
Sales PC 1	B-Sales-PC1	Fa0	192.168.5.30	255.255.255.0	192.168.5.1
Sales PC 2	B-Sales-PC2	Fa0	192.168.5.31	255.255.255.0	192.168.5.1
Sales PC 3	B-Sales-PC3	Fa0	192.168.5.32	255.255.255.0	192.168.5.1
Mktg PC 1	B-Mktg-PC1	Fa0	192.168.5.40	255.255.255.0	192.168.5.1
Mktg PC 2	B-Mktg-PC2	Fa0	192.168.5.41	255.255.255.0	192.168.5.1
Mktg PC 3	B-Mktg-PC3	Fa0	192.168.5.42	255.255.255.0	192.168.5.1
Mgmt PC	B-Mgmt-PC	Fa0	192.168.5.20	255.255.255.0	192.168.5.1

Table 1: Comprehensive Branch 2 IP Addressing Plan

2.2 Physical Connection Matrix

#	From	Port	To	Port	Type
1	R3	Se0/0/1 (DCE)	R4	Se0/0/0 (DTE)	Serial DCE
2	R4	Gi0/0	CS	Gi0/1	Copper ST
3	CS	Gi0/2	B-IT-SW	Gi0/1	Copper ST
4	CS	Fa0/1	B2-AAA	Fa0	Copper ST
5	CS	Fa0/2	B2-Log	Fa0	Copper ST
6	CS	Fa0/3	B-Emp-SW	Gi0/1	Copper ST
7	B-IT-SW	Fa0/1	B-IT-PC1	Fa0	Copper ST
8	B-IT-SW	Fa0/2	B-IT-PC2	Fa0	Copper ST
9	B-IT-SW	Fa0/3	B-IT-PC3	Fa0	Copper ST
10	B-Emp-SW	Fa0/1	B-Sales-PC1	Fa0	Copper ST
11	B-Emp-SW	Fa0/2	B-Sales-PC2	Fa0	Copper ST
12	B-Emp-SW	Fa0/3	B-Sales-PC3	Fa0	Copper ST
13	B-Emp-SW	Fa0/4	B-Mktg-PC1	Fa0	Copper ST
14	B-Emp-SW	Fa0/5	B-Mktg-PC2	Fa0	Copper ST
15	B-Emp-SW	Fa0/6	B-Mktg-PC3	Fa0	Copper ST
16	B-Emp-SW	Fa0/7	B-Mgmt-PC	Fa0	Copper ST

Table 2: Full Branch Connectivity Matrix

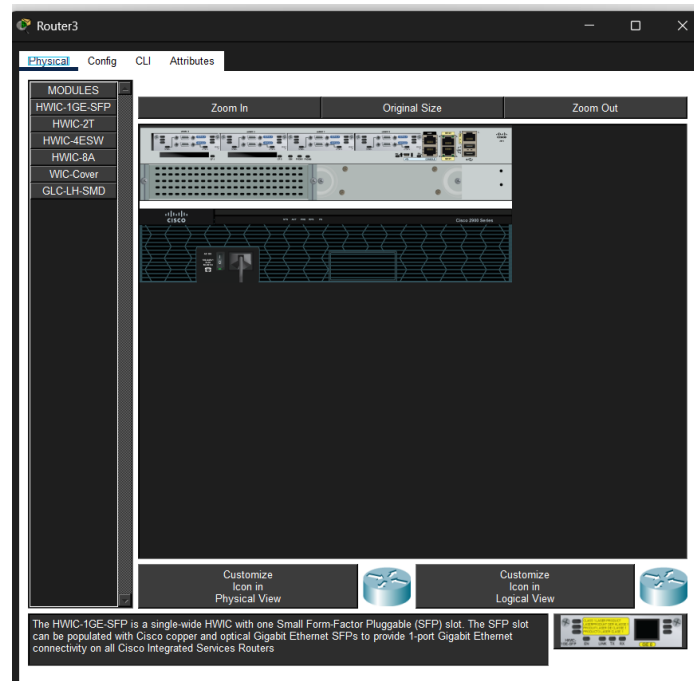


Figure 2: Router 3 Physical HWIC-2T Module Installation.

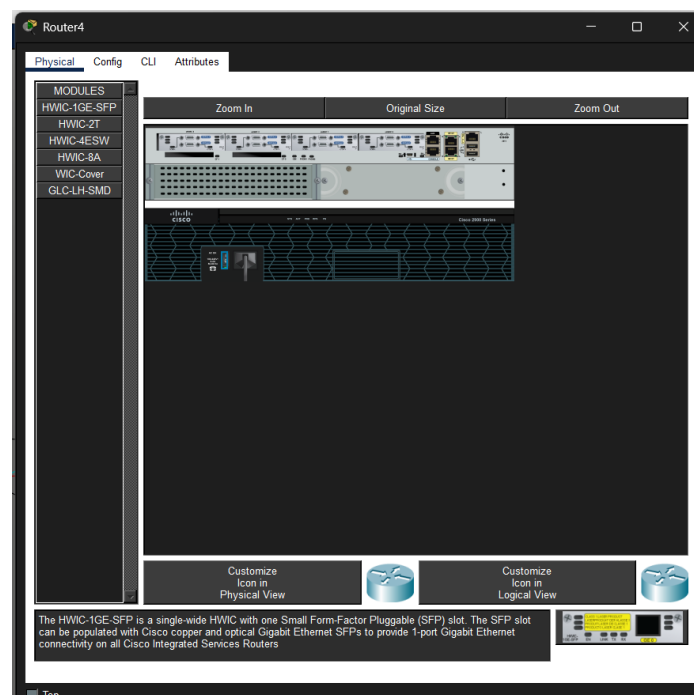


Figure 3: Router 4 Physical HWIC-2T Module Installation.

4 Phase 2: Core & Switching Configuration

Step 3 – R4 Edge Router Initialization

R4 was configured with management security, SSH access, and IP addressing for both the LAN and the WAN serial interface.

CLI: R4 Basic Config

```
Router> enable
Router# configure terminal
R4(config)# hostname R4
R4(config)# service password-encryption
R4(config)# enable secret NewBranch@55
R4(config)# username Admin privilege 15 secret adminP@55
R4(config)# line console 0
R4(config-line)# login local
R4(config-line)# exit
R4(config)# interface GigabitEthernet0/0
R4(config-if)# ip address 192.168.5.1 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
```

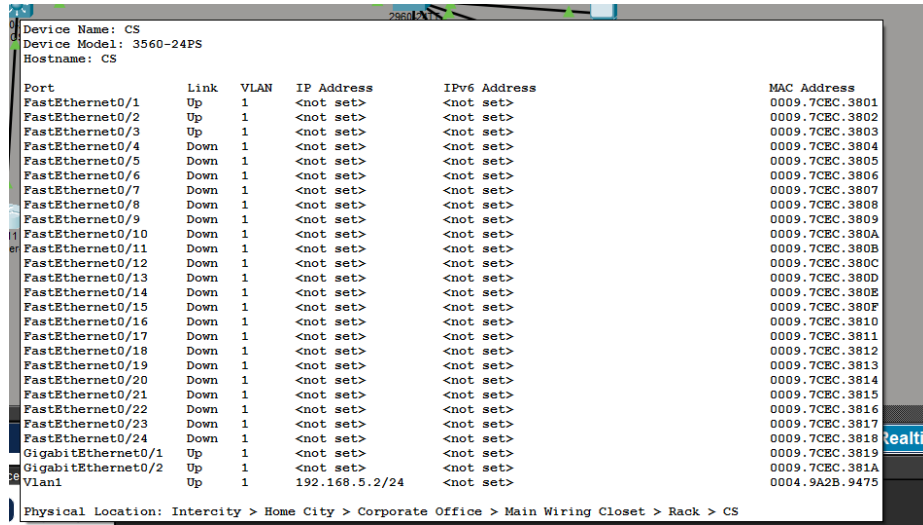
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	192.168.5.1/24	<not set>	0001.96D7.8C01
GigabitEthernet0/1	Down	--	<not set>	<not set>	0001.96D7.8C02
GigabitEthernet0/2	Down	--	<not set>	<not set>	0001.96D7.8C03
Serial0/0/0	Up	--	172.16.3.254/24	<not set>	<not set>
Serial0/0/1	Down	--	<not set>	<not set>	<not set>
Serial0/1/0	Down	--	<not set>	<not set>	<not set>
Serial0/1/1	Down	--	<not set>	<not set>	<not set>
Serial0/2/0	Down	--	<not set>	<not set>	<not set>
Serial0/2/1	Down	--	<not set>	<not set>	<not set>
Serial0/3/0	Down	--	<not set>	<not set>	<not set>
Serial0/3/1	Down	--	<not set>	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0060.7070.8858

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router4

Figure 4: R4 Interface and IP Status Verification.

Step 4 – Core Switch (CS) Deployment

The 3560 switch serves as the backbone for the branch. It was assigned a management SVI on VLAN 1.



The screenshot shows a network device configuration page. At the top, it displays 'Device Name: CS', 'Device Model: 3560-24PS', and 'Hostname: CS'. Below this is a table with columns: Port, Link, VLAN, IP Address, IPv6 Address, and MAC Address. The table lists 24 FastEthernet ports (FastEthernet0/1 to FastEthernet0/24) and two GigabitEthernet ports (GigabitEthernet0/1 and GigabitEthernet0/2). Most ports are 'Down' and have 'VLAN 1' and 'IP Address <not set>'. The MAC addresses for these ports range from 0009.7CEC.3801 to 0009.7CEC.381A. The GigabitEthernet0/1 port is 'Up' and has 'VLAN 1' and 'IP Address 192.168.5.2/24'. The GigabitEthernet0/2 port is 'Up' and has 'VLAN 1' and 'IP Address <not set>'. At the bottom, the physical location is listed as 'Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > CS'.

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Up	1	<not set>	<not set>	0009.7CEC.3801
FastEthernet0/2	Up	1	<not set>	<not set>	0009.7CEC.3802
FastEthernet0/3	Up	1	<not set>	<not set>	0009.7CEC.3803
FastEthernet0/4	Down	1	<not set>	<not set>	0009.7CEC.3804
FastEthernet0/5	Down	1	<not set>	<not set>	0009.7CEC.3805
FastEthernet0/6	Down	1	<not set>	<not set>	0009.7CEC.3806
FastEthernet0/7	Down	1	<not set>	<not set>	0009.7CEC.3807
FastEthernet0/8	Down	1	<not set>	<not set>	0009.7CEC.3808
FastEthernet0/9	Down	1	<not set>	<not set>	0009.7CEC.3809
FastEthernet0/10	Down	1	<not set>	<not set>	0009.7CEC.380A
FastEthernet0/11	Down	1	<not set>	<not set>	0009.7CEC.380B
FastEthernet0/12	Down	1	<not set>	<not set>	0009.7CEC.380C
FastEthernet0/13	Down	1	<not set>	<not set>	0009.7CEC.380D
FastEthernet0/14	Down	1	<not set>	<not set>	0009.7CEC.380E
FastEthernet0/15	Down	1	<not set>	<not set>	0009.7CEC.380F
FastEthernet0/16	Down	1	<not set>	<not set>	0009.7CEC.3810
FastEthernet0/17	Down	1	<not set>	<not set>	0009.7CEC.3811
FastEthernet0/18	Down	1	<not set>	<not set>	0009.7CEC.3812
FastEthernet0/19	Down	1	<not set>	<not set>	0009.7CEC.3813
FastEthernet0/20	Down	1	<not set>	<not set>	0009.7CEC.3814
FastEthernet0/21	Down	1	<not set>	<not set>	0009.7CEC.3815
FastEthernet0/22	Down	1	<not set>	<not set>	0009.7CEC.3816
FastEthernet0/23	Down	1	<not set>	<not set>	0009.7CEC.3817
FastEthernet0/24	Down	1	<not set>	<not set>	0009.7CEC.3818
GigabitEthernet0/1	Up	1	<not set>	<not set>	0009.7CEC.3819
GigabitEthernet0/2	Up	1	<not set>	<not set>	0009.7CEC.381A
Vlan1	Up	1	192.168.5.2/24	<not set>	0004.9A2B.9475

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > CS

Figure 5: CS SVI and Interface Status.

5 Phase 2: Core & Switching Configuration

Step 3 – R4 Edge Router Initialization

R4 was configured with management security, SSH v2, and IP addressing. Remote access is restricted to SSH only, with a 5-minute timeout and localized authentication.

CLI: R4 Core Config

```
R4(config)# hostname R4
R4(config)# service password-encryption
R4(config)# enable secret NewBranch@55
R4(config)# username Admin privilege 15 secret adminP@55
R4(config)# ip domain-name company.local
R4(config)# crypto key generate rsa 1024
R4(config)# ip ssh version 2
R4(config)# line vty 0 4
R4(config-line)# transport input ssh
R4(config-line)# exec-timeout 5 0
R4(config-line)# exit
R4(config)# interface GigabitEthernet0/0
R4(config-if)# ip address 192.168.5.1 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# interface Serial0/0/0
R4(config-if)# ip address 172.16.3.254 255.255.255.0
R4(config-if)# no shutdown
```

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	192.168.5.1/24	<not set>	0001.96D7.8C01
GigabitEthernet0/1	Down	--	<not set>	<not set>	0001.96D7.8C02
GigabitEthernet0/2	Down	--	<not set>	<not set>	0001.96D7.8C03
Serial0/0/0	Up	--	172.16.3.254/24	<not set>	<not set>
Serial0/0/1	Down	--	<not set>	<not set>	<not set>
Serial0/1/0	Down	--	<not set>	<not set>	<not set>
Serial0/1/1	Down	--	<not set>	<not set>	<not set>
Serial0/2/0	Down	--	<not set>	<not set>	<not set>
Serial0/2/1	Down	--	<not set>	<not set>	<not set>
Serial0/3/0	Down	--	<not set>	<not set>	<not set>
Serial0/3/1	Down	--	<not set>	<not set>	<not set>
Vlan1	Down	1	<not set>	<not set>	0060.7070.8858

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router4

Figure 6: R4 Interface and IP Status Verification.

Step 4 – ISP Router (R3) Update

To enable communication with the new branch, the ISP router R3 was updated with the new serial interface and OSPF area 0 advertisements.

CLI: R3 Serial Update

```

R3(config)# interface Serial0/0/1
R3(config-if)# ip address 172.16.3.253 255.255.255.0
R3(config-if)# clock rate 128000
R3(config-if)# no shutdown
R3(config)# router ospf 1
R3(config-router)# network 172.16.3.0 0.0.0.255 area 0

```

Step 5 – Switching Infrastructure (CS & Access)

The Core Switch (CS) and departmental access switches (B-IT-SW, B-Emp-SW) were configured with management SVIs and SSH.

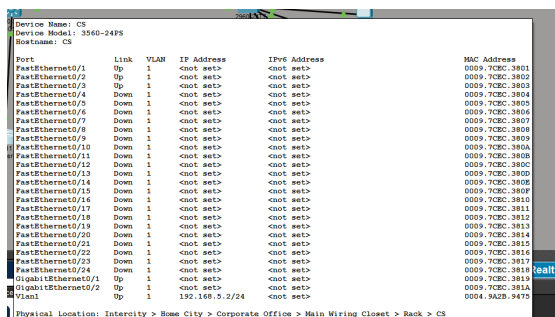
CLI: Switching Management

```

! Core Switch (CS)
CS(config)# interface Vlan1
CS(config-if)# ip address 192.168.5.2 255.255.255.0
CS(config-if)# no shutdown
CS(config)# ip default-gateway 192.168.5.1

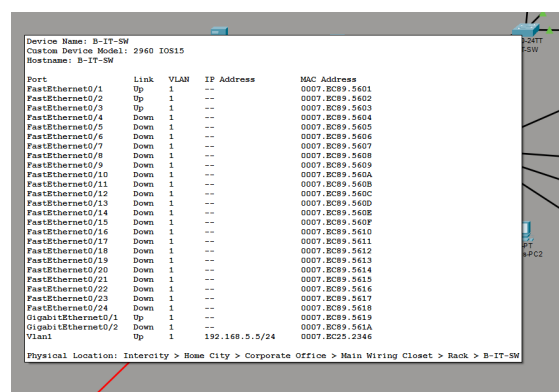
! Access Switches (SVI for management)
B-IT-SW(config)# interface Vlan1
B-IT-SW(config-if)# ip address 192.168.5.5 255.255.255.0
B-IT-SW(config-if)# no shutdown

```



Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Up	1	<not set>	<not set>	0009.7C8C.3801
FastEthernet0/2	Up	1	<not set>	<not set>	0009.7C8C.3802
FastEthernet0/3	Up	1	<not set>	<not set>	0009.7C8C.3803
FastEthernet0/4	Down	1	<not set>	<not set>	0009.7C8C.3804
FastEthernet0/5	Down	1	<not set>	<not set>	0009.7C8C.3805
FastEthernet0/6	Down	1	<not set>	<not set>	0009.7C8C.3806
FastEthernet0/7	Down	1	<not set>	<not set>	0009.7C8C.3807
FastEthernet0/8	Down	1	<not set>	<not set>	0009.7C8C.3808
FastEthernet0/9	Down	1	<not set>	<not set>	0009.7C8C.3809
FastEthernet0/10	Down	1	<not set>	<not set>	0009.7C8C.380A
FastEthernet0/11	Down	1	<not set>	<not set>	0009.7C8C.380B
FastEthernet0/12	Down	1	<not set>	<not set>	0009.7C8C.380C
FastEthernet0/13	Down	1	<not set>	<not set>	0009.7C8C.380D
FastEthernet0/14	Down	1	<not set>	<not set>	0009.7C8C.380E
FastEthernet0/15	Down	1	<not set>	<not set>	0009.7C8C.380F
FastEthernet0/16	Down	1	<not set>	<not set>	0009.7C8C.3810
FastEthernet0/17	Down	1	<not set>	<not set>	0009.7C8C.3811
FastEthernet0/18	Down	1	<not set>	<not set>	0009.7C8C.3812
FastEthernet0/19	Down	1	<not set>	<not set>	0009.7C8C.3813
FastEthernet0/20	Down	1	<not set>	<not set>	0009.7C8C.3814
FastEthernet0/21	Down	1	<not set>	<not set>	0009.7C8C.3815
FastEthernet0/22	Down	1	<not set>	<not set>	0009.7C8C.3816
FastEthernet0/23	Down	1	<not set>	<not set>	0009.7C8C.3817
FastEthernet0/24	Down	1	<not set>	<not set>	0009.7C8C.3818
GigabitEthernet0/1	Up	1	<not set>	<not set>	0009.7C8C.3819
GigabitEthernet0/2	Up	1	<not set>	<not set>	0009.7C8C.381A
Vlan1	Up	1	192.168.5.2/24	<not set>	0004.9A2B.9475

Figure 7: CS SVI Status.



Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0007.BC89.5601
FastEthernet0/2	Up	1	--	0007.BC89.5602
FastEthernet0/3	Up	1	--	0007.BC89.5603
FastEthernet0/4	Down	1	--	0007.BC89.5604
FastEthernet0/5	Down	1	--	0007.BC89.5605
FastEthernet0/6	Down	1	--	0007.BC89.5606
FastEthernet0/7	Down	1	--	0007.BC89.5607
FastEthernet0/8	Down	1	--	0007.BC89.5608
FastEthernet0/9	Down	1	--	0007.BC89.5609
FastEthernet0/10	Down	1	--	0007.BC89.560A
FastEthernet0/11	Down	1	--	0007.BC89.560B
FastEthernet0/12	Down	1	--	0007.BC89.560C
FastEthernet0/13	Down	1	--	0007.BC89.560D
FastEthernet0/14	Down	1	--	0007.BC89.560E
FastEthernet0/15	Down	1	--	0007.BC89.560F
FastEthernet0/16	Down	1	--	0007.BC89.5610
FastEthernet0/17	Down	1	--	0007.BC89.5611
FastEthernet0/18	Down	1	--	0007.BC89.5612
FastEthernet0/19	Down	1	--	0007.BC89.5613
FastEthernet0/20	Down	1	--	0007.BC89.5614
FastEthernet0/21	Down	1	--	0007.BC89.5615
FastEthernet0/22	Down	1	--	0007.BC89.5616
FastEthernet0/23	Down	1	--	0007.BC89.5617
FastEthernet0/24	Down	1	--	0007.BC89.5618
GigabitEthernet0/1	Up	1	--	0007.BC89.5619
GigabitEthernet0/2	Down	1	--	0007.BC89.561A
Vlan1	Up	1	192.168.5.5/24	0007.BC25.2346

Figure 8: B-IT-SW Management Link.

6 Phase 3: Centralized Security Services

Step 6 – AAA & Logging CLI Implementation

All network devices were configured to point to the centralized AAA and SYSLOG servers for unified management.

CLI: AAA Logging CLI

```
! AAA Configuration
R4(config)# aaa new-model
R4(config)# tacacs-server host 192.168.5.3 key TacacsKey123
R4(config)# aaa authentication login default group tacacs+
local
R4(config)# aaa authorization exec default group tacacs+
local

! Logging Configuration
R4(config)# logging host 192.168.5.4
R4(config)# logging trap notifications
```

Step 7 – Security Server GUI Configuration

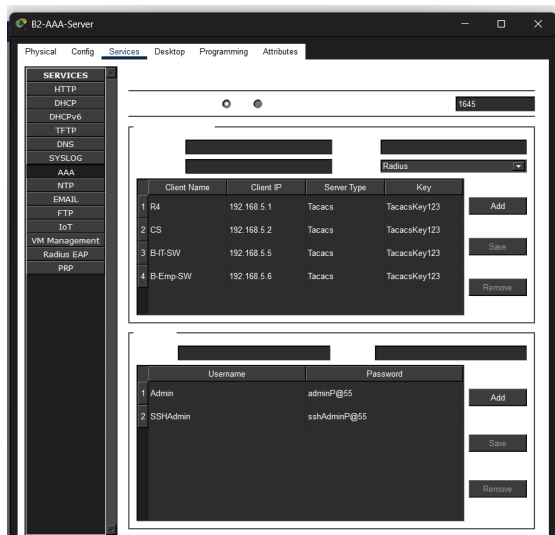


Figure 9: AAA Tacacs Service Setup.

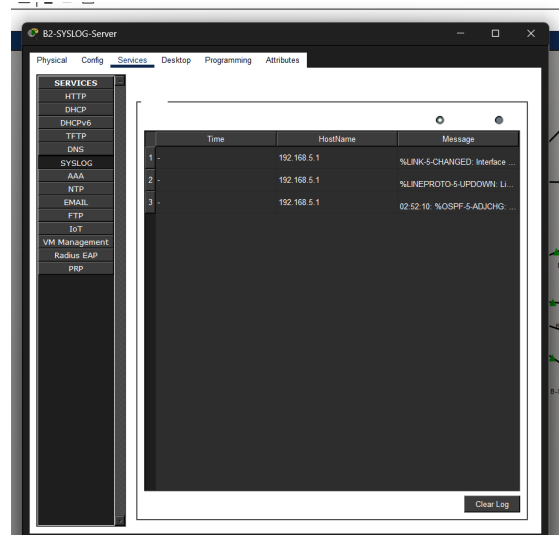


Figure 10: Syslog Aggregate View.

7 Phase 4: Routing & Connectivity Verification

Step 8 – OSPF Convergence & Banners

Routing was finalized with OSPF for the internal LAN, ensuring all routes are dynamically learned without auto-summarization to maintain prefix integrity.

CLI: OSPF Banner Config

```
R4(config)# router ospf 1
R4(config-router)# network 192.168.5.0 0.0.0.255 area 0
R4(config-router)# network 172.16.3.0 0.0.0.255 area 0
R4(config-router)# passive-interface GigabitEthernet0/0
R4(config-router)# no auto-summary
R4(config-router)# exit
R4(config)# banner motd ^C
*****
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED BY LAW.
ALL ACTIVITIES ARE MONITORED BY NETWORK SECURITY TEAM.
*****
^C
```

Step 9 – Host & Server Provisioning Verification

Every endpoint in the new branch expansion was verified for IP reachability and service status. This includes the centralized security servers (AAA and SYSLOG) as well as the departmental PC pools.



Figure 11: B2-AAA Host Status.

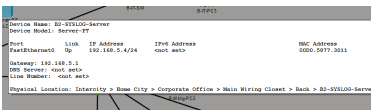


Figure 12: B2-Log Host Status.

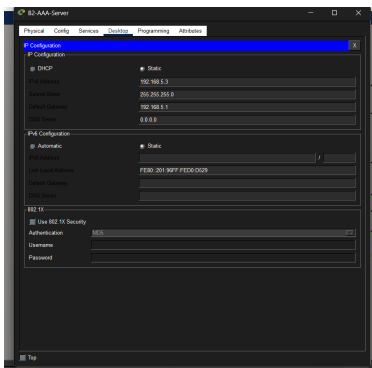


Figure 13: AAA Desktop Config.

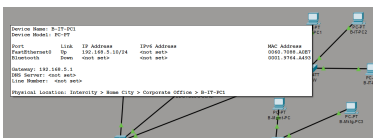


Figure 14: IT-PC1 Status.

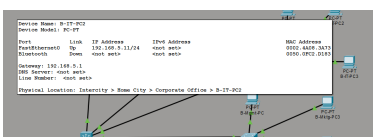


Figure 15: IT-PC2 Status.

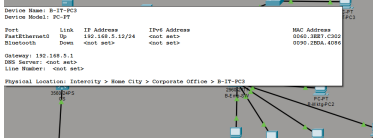


Figure 16: IT-PC3 Status.

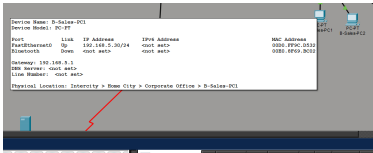


Figure 17: Sales-PC1.

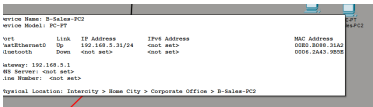


Figure 18: Sales-PC2.



Figure 19: Sales-PC3.

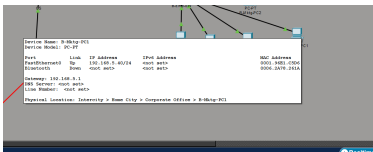


Figure 20: Marketing-PC1.

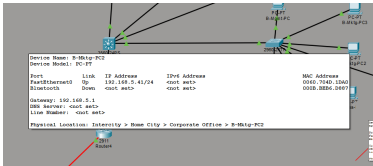


Figure 21: Marketing-PC2.

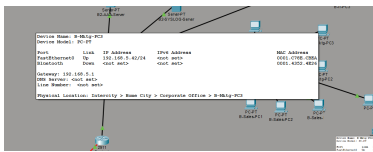


Figure 22: Marketing-PC3.

8 Phase 5: Advanced Security Policies

Step 10 – Site-to-Site IPsec VPN

To secure traffic between the Main Branch (R2) and the New Branch (R4), a Site-to-Site VPN was implemented. This tunnel ensures that all cross-branch communication remains confidential.

CLI: R4 VPN Implementation

```
R4(config)# crypto isakmp policy 10
R4(config-isakmp)# encryption aes
R4(config-isakmp)# hash sha
R4(config-isakmp)# authentication pre-share
R4(config-isakmp)# group 2
R4(config-isakmp)# exit
R4(config)# crypto isakmp key VPN@55 address 172.16.1.2
R4(config)# crypto ipsec transform-set MYSET esp-aes esp-
sha-hmac
R4(config)# access-list 110 permit ip 192.168.5.0 0.0.0.255
192.168.1.0 0.0.0.255
R4(config)# crypto map MYMAP 10 ipsec-isakmp
R4(config-crypto-map)# set peer 172.16.1.2
R4(config-crypto-map)# set transform-set MYSET
R4(config-crypto-map)# match address 110
R4(config)# interface Serial0/0/0
R4(config-if)# crypto map MYMAP
```

Step 11 – Policy Enforcement (Brute-Force & Port Security)

CLI: Security Hardening

```

! Brute-Force Protection
R4(config)# login block-for 120 attempts 3 within 60

! Port Security (Access Switches)
Switch(config)# interface range fa0/1 - 24
Switch(config-if-range)# switchport port-security
Switch(config-if-range)# switchport port-security mac-
address sticky
Switch(config-if-range)# switchport port-security violation
restrict

```

Device Name: B-EMP-SW
Custom Device Model: 2960 IOS15
Hostname: B-EMP-SW

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0090.2143.6601
FastEthernet0/2	Up	1	--	0090.2143.6602
FastEthernet0/3	Up	1	--	0090.2143.6603
FastEthernet0/4	Up	1	--	0090.2143.6604
FastEthernet0/5	Up	1	--	0090.2143.6605
FastEthernet0/6	Up	1	--	0090.2143.6606
FastEthernet0/7	Up	1	--	0090.2143.6607
FastEthernet0/8	Up	1	--	0090.2143.6608
FastEthernet0/9	Down	1	--	0090.2143.6609
FastEthernet0/10	Down	1	--	0090.2143.660A
FastEthernet0/11	Down	1	--	0090.2143.660B
FastEthernet0/12	Down	1	--	0090.2143.660C
FastEthernet0/13	Down	1	--	0090.2143.660D
FastEthernet0/14	Down	1	--	0090.2143.660E
FastEthernet0/15	Down	1	--	0090.2143.660F
FastEthernet0/16	Down	1	--	0090.2143.6610
FastEthernet0/17	Down	1	--	0090.2143.6611
FastEthernet0/18	Down	1	--	0090.2143.6612
FastEthernet0/19	Down	1	--	0090.2143.6613
FastEthernet0/20	Down	1	--	0090.2143.6614
FastEthernet0/21	Down	1	--	0090.2143.6615
FastEthernet0/22	Down	1	--	0090.2143.6616
FastEthernet0/23	Down	1	--	0090.2143.6617
FastEthernet0/24	Down	1	--	0090.2143.6618
GigabitEthernet0/1	Down	1	--	0090.2143.6619
GigabitEthernet0/2	Down	1	--	0090.2143.661A
Vlan1	Up	1	192.168.5.6/24	0030.F221.9CA2

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > B-EMP-SW

Figure 23: Port Security Sticky MACs.

Device Name: B-IT-SW
Custom Device Model: 2960 IOS15
Hostname: B-IT-SW

Port	Link	VLAN	IP Address	MAC Address
FastEthernet0/1	Up	1	--	0007.EC89.5601
FastEthernet0/2	Up	1	--	0007.EC89.5602
FastEthernet0/3	Up	1	--	0007.EC89.5603
FastEthernet0/4	Down	1	--	0007.EC89.5604
FastEthernet0/5	Down	1	--	0007.EC89.5605
FastEthernet0/6	Down	1	--	0007.EC89.5606
FastEthernet0/7	Down	1	--	0007.EC89.5607
FastEthernet0/8	Down	1	--	0007.EC89.5608
FastEthernet0/9	Down	1	--	0007.EC89.5609
FastEthernet0/10	Down	1	--	0007.EC89.560A
FastEthernet0/11	Down	1	--	0007.EC89.560B
FastEthernet0/12	Down	1	--	0007.EC89.560C
FastEthernet0/13	Down	1	--	0007.EC89.560D
FastEthernet0/14	Down	1	--	0007.EC89.560E
FastEthernet0/15	Down	1	--	0007.EC89.560F
FastEthernet0/16	Down	1	--	0007.EC89.5610
FastEthernet0/17	Down	1	--	0007.EC89.5611
FastEthernet0/18	Down	1	--	0007.EC89.5612
FastEthernet0/19	Down	1	--	0007.EC89.5613
FastEthernet0/20	Down	1	--	0007.EC89.5614
FastEthernet0/21	Down	1	--	0007.EC89.5615
FastEthernet0/22	Down	1	--	0007.EC89.5616
FastEthernet0/23	Down	1	--	0007.EC89.5617
FastEthernet0/24	Down	1	--	0007.EC89.5618
GigabitEthernet0/1	Up	1	--	0007.EC89.5619
GigabitEthernet0/2	Down	1	--	0007.EC89.561A
Vlan1	Up	1	192.168.5.5/24	0007.EC29.2346

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > B-IT-SW

Figure 24: Hardened Interface Status.

9 Configuration Verification & Evidence

This section provides empirical verification and configuration proofs for all ten security and routing requirements implemented in the new branch. Each configuration is backed by active running configuration parameters, CLI operational outputs, and corresponding Packet Tracer validation screenshots.

9.1 Configuration 1: OSPF Routing Protocol

OSPF routing was configured on the new branch router R4 and the ISP router R3 to dynamically establish network routes and ensure full end-to-end routing convergence between all branches.

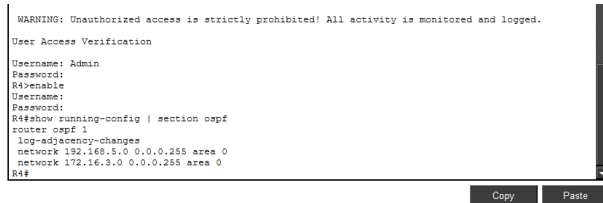


Figure 25: R3 OSPF Configuration & Protocols

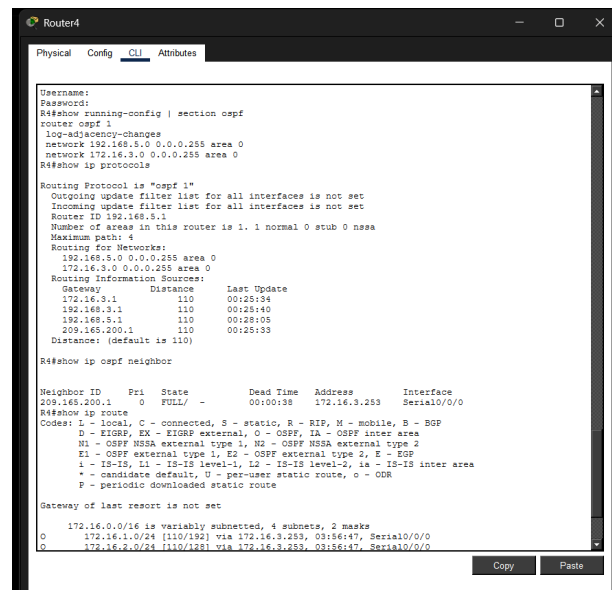


Figure 26: R3 OSPF Neighbors



Figure 27: R3 Routing Table

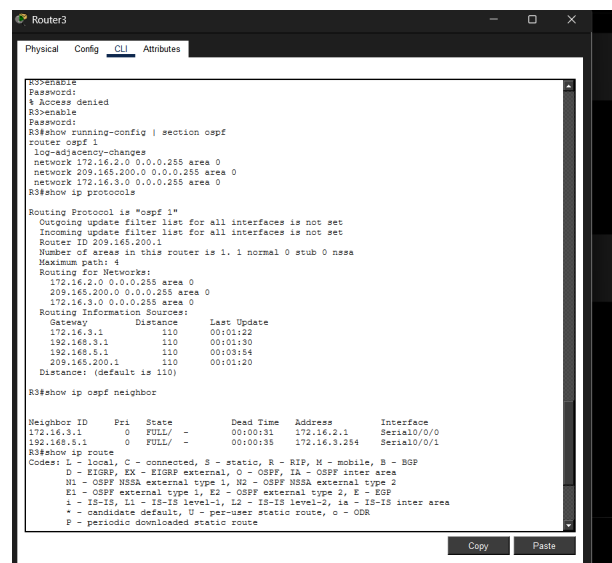


Figure 28: R4 OSPF Neighbor Adjacency

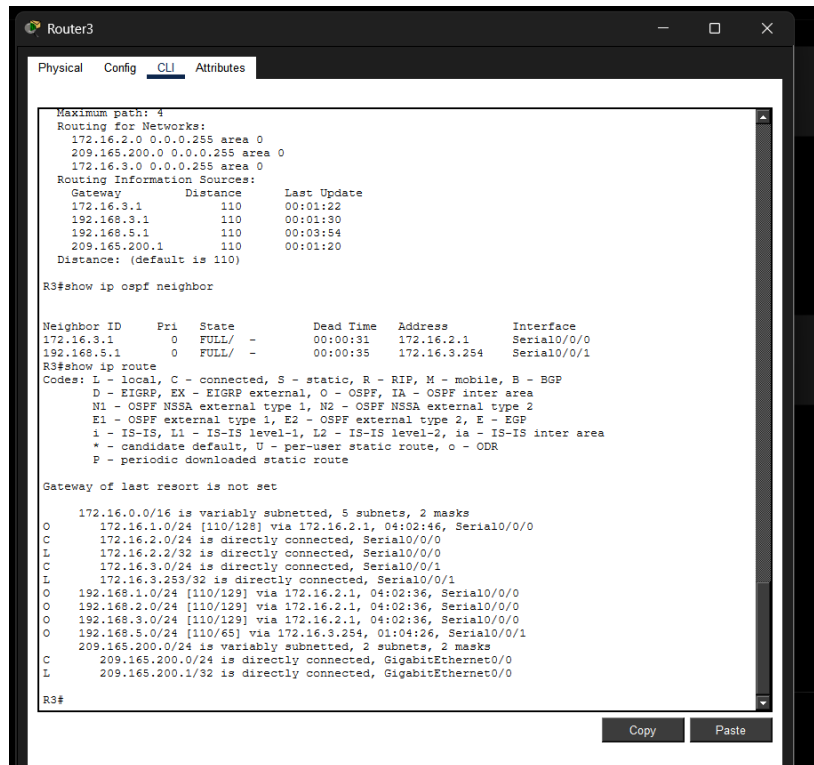


Figure 29: R4 Routing Table Convergence

CLI: OSPF Routing Verification on R3 and R4

```

R3#show running-config | section ospf
router ospf 1
 log-adjacency-changes
 network 172.16.2.0 0.0.0.255 area 0
 network 209.165.200.0 0.0.0.255 area 0
 network 172.16.3.0 0.0.0.255 area 0
R3#show ip protocols
  
```

```

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 209.165.200.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    172.16.2.0 0.0.0.255 area 0
    209.165.200.0 0.0.0.255 area 0
    172.16.3.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.3.1       110           00:01:22
    192.168.3.1      110           00:01:30
    192.168.5.1      110           00:03:54
    209.165.200.1    110           00:01:20
  Distance: (default is 110)
  
```

```
R3#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address
172.16.3.1	0	FULL/ -	00:00:31	172.16.2.1
Serial0/0/0				
192.168.5.1	0	FULL/ -	00:00:35	172.16.3.254
Serial0/0/1				

```
R3#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile
        , B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
          area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
          type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E -
          EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
          IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```

    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O    172.16.1.0/24 [110/128] via 172.16.2.1, 04:02:46,
    Serial0/0/0
C    172.16.2.0/24 is directly connected, Serial0/0/0
L    172.16.2.2/32 is directly connected, Serial0/0/0
C    172.16.3.0/24 is directly connected, Serial0/0/1
L    172.16.3.253/32 is directly connected, Serial0/0/1
O    192.168.1.0/24 [110/129] via 172.16.2.1, 04:02:36, Serial0
    /0/0
O    192.168.2.0/24 [110/129] via 172.16.2.1, 04:02:36, Serial0
    /0/0
O    192.168.3.0/24 [110/129] via 172.16.2.1, 04:02:36, Serial0
    /0/0
O    192.168.5.0/24 [110/65] via 172.16.3.254, 01:04:26, Serial0
    /0/1
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.0/24 is directly connected, GigabitEthernet0
    /0
L    209.165.200.1/32 is directly connected, GigabitEthernet0
    /0
```

```
R4#show running-config | section ospf
router ospf 1
 log-adjacency-changes
```

```

network 192.168.5.0 0.0.0.255 area 0
network 172.16.3.0 0.0.0.255 area 0
R4#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 192.168.5.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.5.0 0.0.0.255 area 0
    172.16.3.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.3.1       110          00:25:34
    192.168.3.1      110          00:25:40
    192.168.5.1      110          00:28:05
    209.165.200.1    110          00:25:33
  Distance: (default is 110)

R4#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address
      Interface
209.165.200.1    0     FULL/  -        00:00:38    172.16.3.253
      Serial0/0/0
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile
      , B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
          area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
          type 2
      E1 - OSPF external type 1, E2 - OSPF external type 2, E -
          EGP
      i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
          IS-IS inter area
      * - candidate default, U - per-user static route, o - ODR
      P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 4 subnets, 2 masks
O       172.16.1.0/24 [110/192] via 172.16.3.253, 03:56:47,
        Serial0/0/0
O       172.16.2.0/24 [110/128] via 172.16.3.253, 03:56:47,
        Serial0/0/0
C       172.16.3.0/24 is directly connected, Serial0/0/0

```

```
L      172.16.3.254/32 is directly connected, Serial0/0/0
O      192.168.1.0/24 [110/193] via 172.16.3.253, 03:56:47,
Serial0/0/0
O      192.168.2.0/24 [110/193] via 172.16.3.253, 03:56:47,
Serial0/0/0
O      192.168.3.0/24 [110/193] via 172.16.3.253, 03:56:47,
Serial0/0/0
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.5.0/24 is directly connected, GigabitEthernet0/0
L      192.168.5.1/32 is directly connected, GigabitEthernet0/0
O      209.165.200.0/24 [110/65] via 172.16.3.253, 03:56:47,
Serial0/0/0
```

9.2 Configuration 2: Site-to-Site VPN

A secure IPsec VPN tunnel was configured between the original branch edge router R2 and the new branch edge router R4. This ensures that all data transmitted between the two offices is fully encrypted and authenticated.

```

Router4
Physical Config CLI Attributes
CLI
User Access Verification
Username: Admin
Password:
R4>enable
Username:
Password:
R4#show running-config | section crypto
crypto isakmp policy 10
encr aes
authentication pre-share
group 2
crypto isakmp key VPNKey456 address 172.16.2.1
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
set peer 172.16.2.1
set transform-set VPN-SET
match address VPN-TRAFFIC-R4
crypto map VPN-MAP
R4#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
% Invalid input detected at '^' marker.
R4#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
% Invalid input detected at '^' marker.
R4#show access-lists
Extended IP access list sl_def_acl
0 deny tcp any any eq telnet
0 deny tcp any any eq www
0 deny tcp any any eq 22
0 permit tcp any any eq 22
Extended IP access list VPN-TRAFFIC-R4
10 permit ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
20 permit ip 192.168.5.0 0.0.0.255 192.168.2.0 0.0.0.255 (1 match(es))
30 permit ip 192.168.5.0 0.0.0.255 192.168.3.0 0.0.0.255 (1 match(es))
R4#show crypto ipsec sa

```

Figure 30: R2 IPsec VPN Tunnel Status

```

Router4
Physical Config CLI Attributes
CLI
R4#show access-lists
Extended IP access list sl_def_acl
0 deny tcp any any eq telnet
0 deny tcp any any eq www
0 deny tcp any any eq 22
0 permit tcp any any eq 22
Extended IP access list VPN-TRAFFIC-R4
10 permit ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
20 permit ip 192.168.5.0 0.0.0.255 192.168.2.0 0.0.0.255 (1 match(es))
30 permit ip 192.168.5.0 0.0.0.255 192.168.3.0 0.0.0.255 (1 match(es))
R4#show crypto ipsec sa
interface: Serial0/0/0
Crypto map tag: VPN-MAP, local addr 172.16.3.254
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.5.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
current_peer 172.16.2.1 port 500
PERMIT, flags(origin_is_acl):
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
local crypto endpt.: 172.16.3.254, remote crypto endpt.: 172.16.2.1
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
current outbound spi: 0x0(0)
inbound esp sas:
inbound ah sas:
inbound pop sas:
outbound esp sas:
outbound ah sas:
outbound pop sas:
--More--

```

Figure 31: R4 IPsec VPN Tunnel Status

CLI: IPsec VPN Verification on R2 and R4

```

R2#show running-config | section crypto
crypto isakmp policy 10
encr aes
authentication pre-share
group 2
crypto isakmp key VPNKey456 address 172.16.3.254
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
set peer 172.16.3.254
set transform-set VPN-SET
match address VPN-TRAFFIC-R2
crypto map VPN-MAP
R2#show access-lists
Extended IP access list sl_def_acl
0 deny tcp any any eq telnet
0 deny tcp any any eq www
0 deny tcp any any eq 22
0 permit tcp any any eq 22
Extended IP access list VPN-TRAFFIC-R2
10 permit ip 192.168.1.0 0.0.0.255 192.168.5.0 0.0.0.255
20 permit ip 192.168.2.0 0.0.0.255 192.168.5.0 0.0.0.255
30 permit ip 192.168.3.0 0.0.0.255 192.168.5.0 0.0.0.255
R2#show crypto ipsec sa

```

```
interface: Serial0/0/1
  Crypto map tag: VPN-MAP, local addr 172.16.2.1

  protected vrf: (none)
  local ident (addr/mask/prot/port):
    (192.168.1.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port):
    (192.168.5.0/255.255.255.0/0/0)
  current_peer 172.16.3.254 port 500
    PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 172.16.2.1, remote crypto endpt.:
    172.16.3.254
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/1
  current outbound spi: 0x0(0)

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

R4#show running-config | section crypto
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
crypto isakmp key VPNKey456 address 172.16.2.1
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
  set peer 172.16.2.1
  set transform-set VPN-SET
  match address VPN-TRAFFIC-R4
crypto map VPN-MAP
R4#show access-lists
Extended IP access list sl_def_acl
  0 deny tcp any any eq telnet
  0 deny tcp any any eq www
  0 deny tcp any any eq 22
  0 permit tcp any any eq 22
```

```
Extended IP access list VPN-TRAFFIC-R4
 10 permit ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
 20 permit ip 192.168.5.0 0.0.0.255 192.168.2.0 0.0.0.255 (1
    match(es))
 30 permit ip 192.168.5.0 0.0.0.255 192.168.3.0 0.0.0.255 (1
    match(es))

R4#show crypto ipsec sa

interface: Serial0/0/0
  Crypto map tag: VPN-MAP, local addr 172.16.3.254

  protected vrf: (none)
  local  ident (addr/mask/prot/port):
    (192.168.5.0/255.255.255.0/0/0)
  remote  ident (addr/mask/prot/port):
    (192.168.1.0/255.255.255.0/0/0)
  current_peer 172.16.2.1 port 500
    PERMIT, flags={origin_is_acl,}
  #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
  #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

    local crypto endpt.: 172.16.3.254, remote crypto endpt
      .:172.16.2.1
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
  current outbound spi: 0x0(0)

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

  outbound pcp sas:
```

9.3 Configuration 3: AAA TACACS+ Server Authentication

A centralized AAA server using TACACS+ protocol was deployed at IP '192.168.5.3' to control, authenticate, and authorize management login sessions.



Figure 32: AAA TACACS+ Server GUI Setup

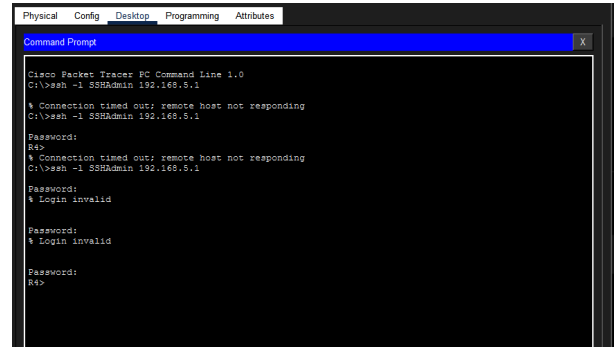


Figure 33: AAA SSH Authentication Success

CLI: AAA TACACS+ Verification on R4

```

R4#show running-config | include tacacs
aaa authentication login default group tacacs+ local
aaa authentication enable default group tacacs+ enable
tacacs-server host 192.168.5.3 key TacacsKey123
R4#show aaa sessions
Total sessions since last reload: 19
Session Id:10
  Unique Id:1
  User Name:Admin
  IP Address:0.0.0.0
  Idle Time: 0
  CT Call Handle: 0

! SSH Authentication and AAA Login Verification via PC CLI:
C:\>ssh -l SSHAdmin 192.168.5.1
Password:
R4>
% Connection timed out; remote host not responding
C:\>ssh -l SSHAdmin 192.168.5.1
Password:
% Login invalid

Password:
% Login invalid
  
```

Password:

R4>

! Router AAA State Post SSH Login:

R4#show running-config | include tacacs

aaa authentication login default group tacacs+ local

aaa authentication enable default group tacacs+ enable

tacacs-server host 192.168.5.3 key TacacsKey123

R4#show aaa sessions

Total sessions since last reload: 19

Session Id:10

Unique Id:1

User Name:Admin

IP Address:0.0.0.0

Idle Time: 0

CT Call Handle: 0

9.4 Configuration 4: Centralized SYSLOG Logging

The SYSLOG server was configured at IP '192.168.5.4' to aggregate log messages from R4 and other new branch network devices.

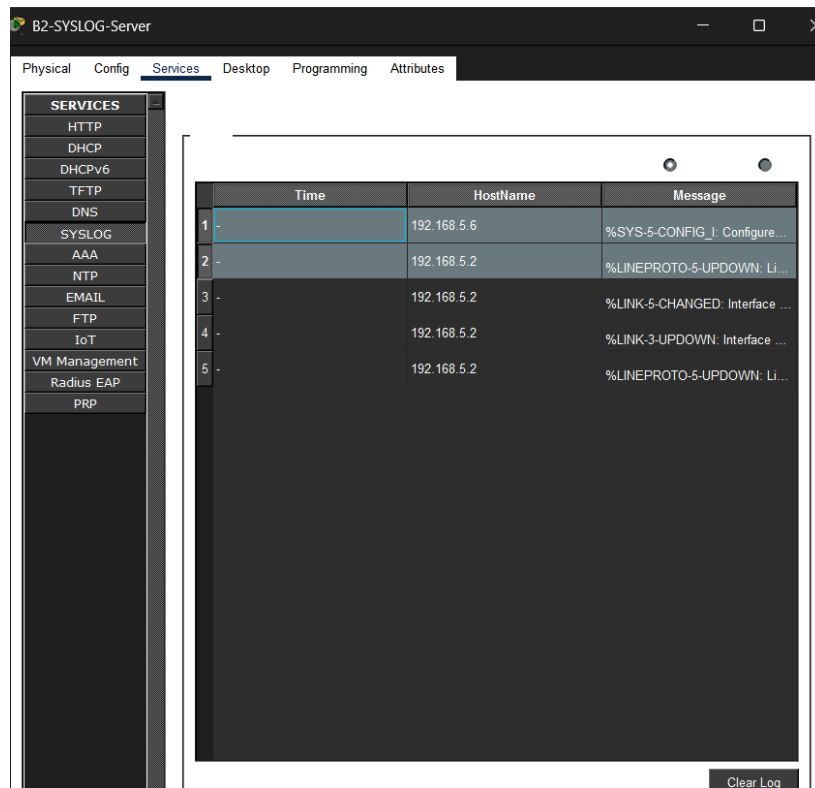


Figure 34: Centralized SYSLOG Server Active Status

CLI: SYSLOG Logging Verification on R4

```
R4#show running-config | include logging
logging 192.168.5.4
```

9.5 Configuration 5: Router Brute-Force Prevention

A security lockout rule was implemented on R4 to prevent brute-force SSH/Console password attacks. Logins are blocked for five minutes (300 seconds) if three invalid password attempts occur within a one-minute (60 seconds) window.

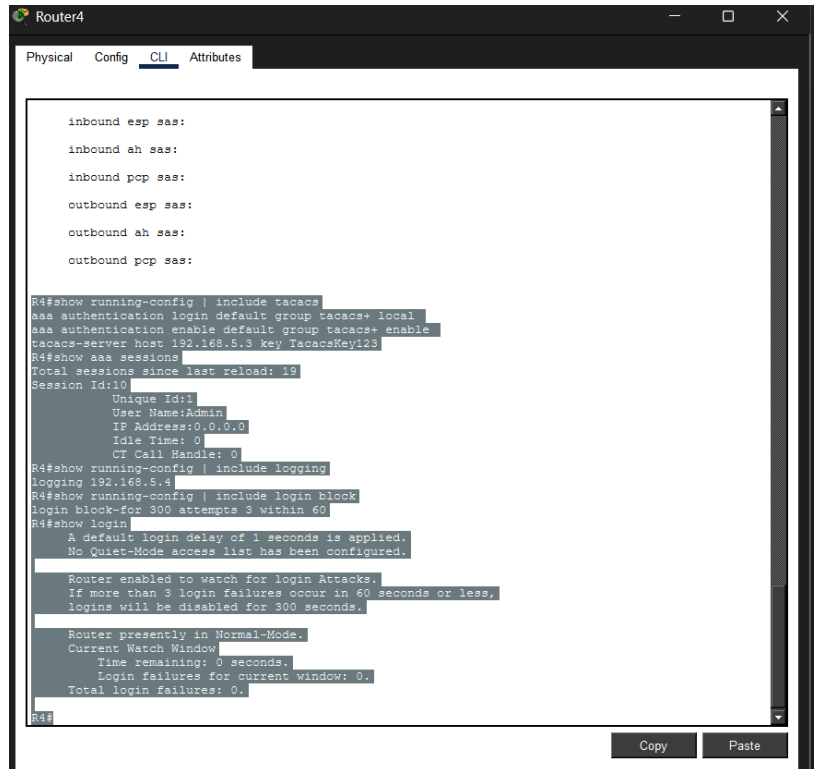


Figure 35: Brute-Force Lockout Monitoring Status

CLI: Brute-Force Lockout Verification on R4

```

R4#show running-config | include login block
login block-for 300 attempts 3 within 60
R4#show login
  A default login delay of 1 seconds is applied.
  No Quiet-Mode access list has been configured.

  Router enabled to watch for login Attacks.
  If more than 3 login failures occur in 60 seconds or less,
  logins will be disabled for 300 seconds.

  Router presently in Normal-Mode.
  Current Watch Window
    Time remaining: 0 seconds.
    Login failures for current window: 0.
    Total login failures: 0.
  
```

9.6 Configuration 6: Switch Port Security Rules

Port security rules were configured across the switches. Critical infrastructure interfaces (AAA Server, SYSLOG Server, and Management Office PC) are configured with a 'shutdown' violation mode to immediately disable the port on breach. Other devices use 'restrict' mode to drop violating traffic and increment security counters.

```

B-Emp-SW#show run interface Fa0/3
% Invalid input detected at '^' marker.

B-Emp-SW#conf t
Enter configuration commands, one per line. End with CNTL/Z.
B-Emp-SW (config)#interface Fa0/3
B-Emp-SW (config-if)#shutdown
B-Emp-SW (config-if)#no shutdown
B-Emp-SW (config-if)#switchport port-security violation shutdown
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
B-Emp-SW (config-if)#exit
B-Emp-SW (config)#exit
B-Emp-SW#
$SYS-5-CONFIG_I: Configured from console by console

B-Emp-SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
(Count) (Count) (Count)
-----
Fa0/2 1 0 0 Shutdown
Fa0/3 1 1 0 Shutdown
Fa0/4 1 0 0 Restrict
Fa0/5 1 0 0 Restrict
Fa0/6 1 0 0 Restrict
Fa0/7 1 0 0 Restrict
Fa0/8 1 0 0 Restrict
B-Emp-SW#

```

Figure 36: B-Emp-SW Switchport Security Status

```

B-Emp-SW#show port-security interface Fa0/3

WARNING: Unauthorized access is strictly prohibited! All activity is monitored and logged.
User Access Verification
Username: Admin
Password:

B-Emp-SW#enable
B-Emp-SW#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Interface Holdtime Capability Platform Port ID
CS Fa0/1 177
B-Emp-SW#show mac address-table
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1 0001.96d7.8c01 DYNAMIC Fa0/1
1 0007.ec25.2346 DYNAMIC Fa0/1
1 0009.70cc.3803 DYNAMIC Fa0/1
1 00d0.ffff.d532 STATIC Fa0/3
B-Emp-SW#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID Local Interface Holdtime Capability Platform Port ID
CS Fa0/1 136
B-Emp-SW#show mac address-table
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1 0001.96d7.8c01 DYNAMIC Fa0/1
1 0009.70cc.3803 DYNAMIC Fa0/1
1 00d0.ffff.d532 STATIC Fa0/3
B-Emp-SW#show port-security interface Fa0/3

```

Figure 37: B-Emp-SW Port Security Configuration

```

B-IT-SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
(Count) (Count) (Count)
-----
Fa0/1 1 1 0 Restrict
Fa0/2 1 1 0 Restrict
Fa0/3 1 0 0 Restrict
B-IT-SW#

B-IT-SW#show mac address-table
Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
1 0001.96d0.d529 DYNAMIC Gi0/1
1 0001.96d7.8c01 DYNAMIC Gi0/1
1 0002.4a08.5a73 STATIC Fa0/2
1 0009.70cc.381a DYNAMIC Gi0/1
1 0060.7088.a8e7 STATIC Fa0/1
B-IT-SW#

```

Figure 38: B-Emp-SW Active MAC Table

```

B-IT-SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
(Count) (Count) (Count)
-----
Fa0/1 1 1 0 Restrict
Fa0/2 1 1 0 Restrict
Fa0/3 1 0 0 Restrict
B-IT-SW#

B-IT-SW con0 is now available

```

Figure 39: B-IT-SW Port Security Violations

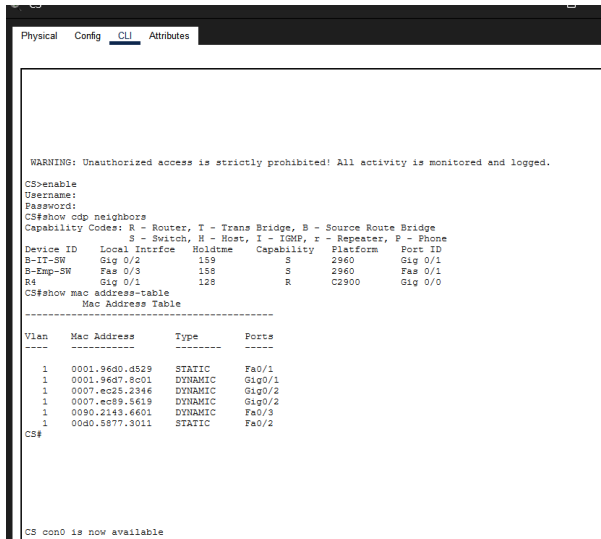


Figure 40: B-IT-SW MAC Address Table

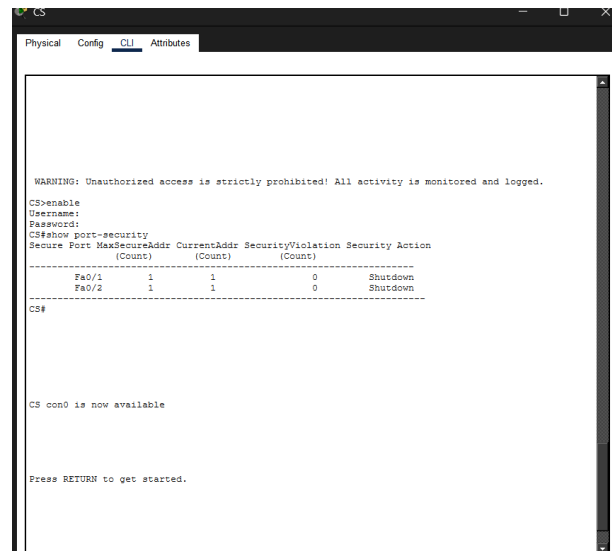


Figure 41: Core Switch (CS) Port Security & Violation Settings

CLI: Port Security Verification on B-Emp-SW, B-IT-SW, and CS

```

! --- B-Emp-SW Port Security Verification \& Shutdown Action ---
B-Emp-SW#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme    Capability  Platform
Port ID
CS              Fas 0/1        177        S           3560
                  Fas 0/3
B-Emp-SW#show mac address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----
1       0001.96d7.8c01   DYNAMIC   Fa0/1
1       0007.ec25.2346   DYNAMIC   Fa0/1
1       0009.7cec.3803   DYNAMIC   Fa0/1
1       00d0.ff9c.d532   STATIC    Fa0/3

B-Emp-SW#show port-security interface fa0/3
Port Security      : Enabled
Port Status        : Secure-up
Violation Mode      : Restrict
Aging Time          : 0 mins
Aging Type          : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1

```

```

Total MAC Addresses      : 1
Configured MAC Addresses : 0
Sticky MAC Addresses     : 1
Last Source Address:Vlan : 00D0.FF9C.D532:1
Security Violation Count : 0

B-Emp-SW#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
B-Emp-SW(config)#interface fa0/3
B-Emp-SW(config-if)#shutdown

B-Emp-SW(config-if)#no shutdown

B-Emp-SW(config-if)#switchport port-security violation shutdown
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to
    administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
    changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,
    changed state to up

B-Emp-SW(config-if)#exit
B-Emp-SW(config)#exit
B-Emp-SW#
%SYS-5-CONFIG_I: Configured from console by console

B-Emp-SW#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security
      Action
              (Count)          (Count)          (Count)
-----
      Fa0/2          1              0              0
      Shutdown
      Fa0/3          1              1              0
      Shutdown
      Fa0/4          1              0              0
      Restrict
      Fa0/5          1              0              0
      Restrict
      Fa0/6          1              0              0
      Restrict
      Fa0/7          1              0              0
      Restrict
      Fa0/8          1              0              0
      Restrict

```

```
! --- B-IT-SW Port Security Verification ---
```

```
B-IT-SW#show port-security
```

```
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security
Action                                     (Count)         (Count)         (Count)
```

```
-----
Fa0/1      1          1          0
Restrict
Fa0/2      1          1          0
Restrict
Fa0/3      1          0          0
Restrict
-----
```

```
B-IT-SW#show cdp neighbors
```

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route
Bridge
```

```
S - Switch, H - Host, I - IGMP, r - Repeater,
P - Phone
```

```
Device ID    Local Intrfce    Holdtme    Capability    Platform
Port ID
CS           Gig 0/1          164
           Gig 0/2          3560
```

```
B-IT-SW#show mac address-table
```

```
Mac Address Table
```

```
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.96d0.d529    DYNAMIC   Gig0/1
1       0001.96d7.8c01    DYNAMIC   Gig0/1
1       0002.4a08.3a73    STATIC    Fa0/2
1       0009.7cec.381a    DYNAMIC   Gig0/1
1       0060.7088.a0e7    STATIC    Fa0/1
-----
```

```
! --- CS (Core Switch) Port Security Verification ---
```

```
CS#show port-security
```

```
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security
Action                                     (Count)         (Count)         (Count)
```

```
-----
Fa0/1      1          1          0
Shutdown
Fa0/2      1          1          0
-----
```

Shutdown

```
CS#show cdp neighbors
```

```
Capability Codes: R - Router, T - Trans Bridge, B - Source Route
                  Bridge
```

```
                  S - Switch, H - Host, I - IGMP, r - Repeater,
                  P - Phone
```

Device ID	Local Intrfce	Holdtme	Capability	Platform
Port ID				
B-IT-SW	Gig 0/2	159	S	2960
	Gig 0/1			
B-Emp-SW	Fas 0/3	158	S	2960
	Fas 0/1			
R4	Gig 0/1	128	R	C2900
	Gig 0/0			

```
CS#show mac address-table
```

```
Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0001.96d0.d529	STATIC	Fa0/1
1	0001.96d7.8c01	DYNAMIC	Gig0/1
1	0007.ec25.2346	DYNAMIC	Gig0/2
1	0007.ec89.5619	DYNAMIC	Gig0/2
1	0090.2143.6601	DYNAMIC	Fa0/3
1	00d0.5877.3011	STATIC	Fa0/2

9.7 Configuration 7: NVRAM Configuration Archival

To prevent config loss upon power cycles or system reboots, the active running configurations of all devices were saved into non-volatile memory (NVRAM) as the startup configuration.

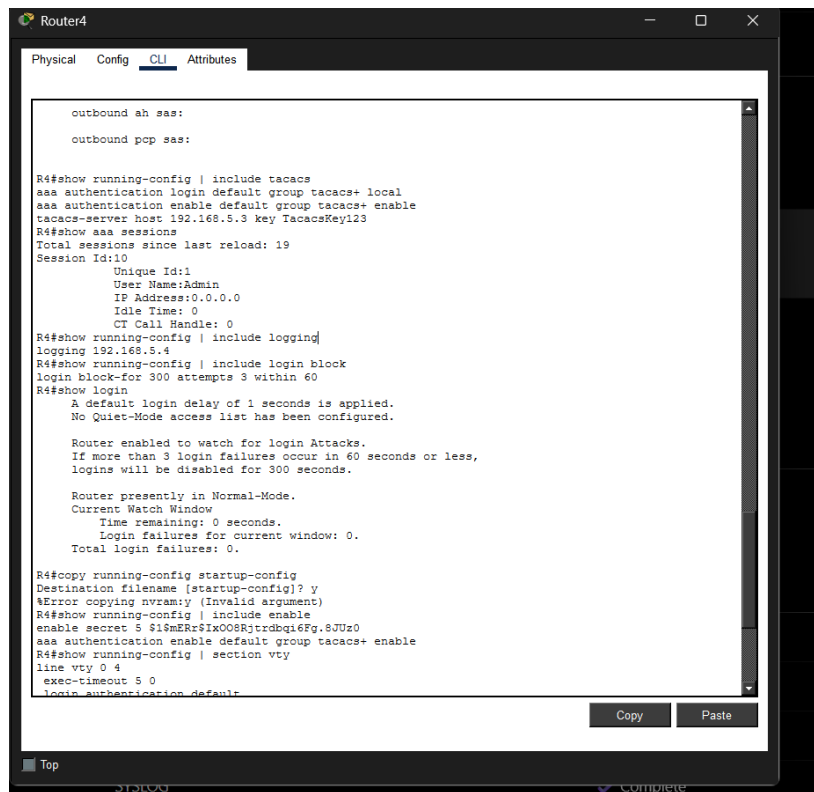


Figure 42: Save Active Configuration to NVRAM

CLI: NVRAM Save Verification on R4

```

R4#copy running-config startup-config
Destination filename [startup-config]? y
%Error copying nvram:y (Invalid argument)
R4#copy running-config startup-config
Destination filename [startup-config]?
[OK]
  
```

9.8 Configuration 8: Secure Enable Secret Implementation

A strong password was established for the privileged EXEC command ('enable secret') on all new branch devices and was automatically hashed to secure the administrator gateway interface.

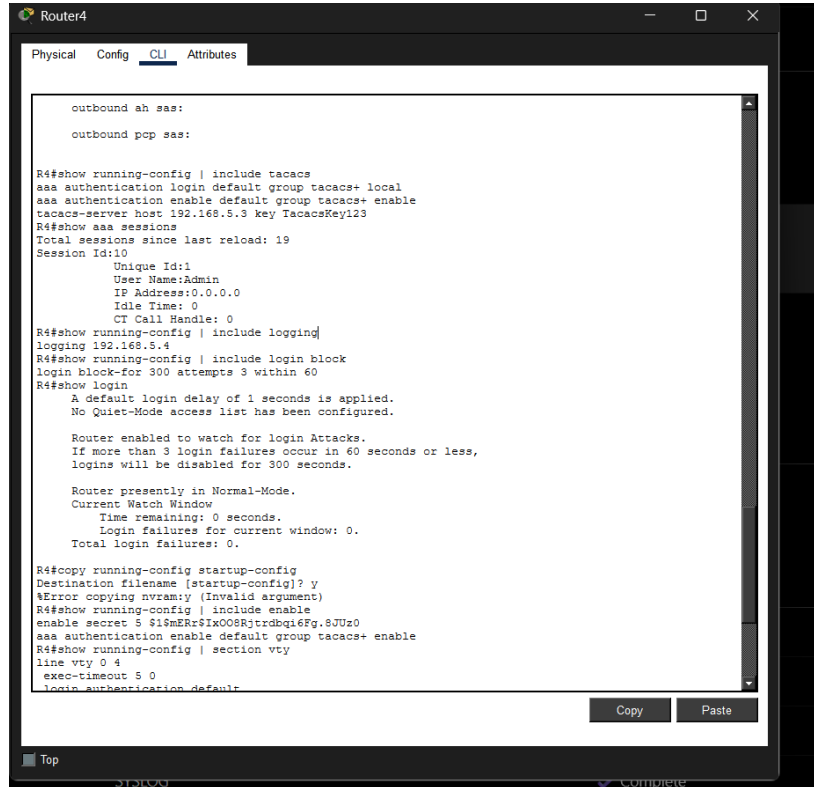


Figure 43: Hashed enable secret Password Verification

CLI: Enable Secret Verification on R4

```

R4#show running-config | include enable
enable secret 5 $1$mERr$Ix008Rjtrdbqi6Fg.8JUz0
aaa authentication enable default group tacacs+ enable
  
```

9.9 Configuration 9: Universal Hashing of Passwords

To prevent unauthorized users from viewing plain-text secrets in switch and router configs, ‘service password-encryption’ was globally enabled. This automatically hashes all configuration-based secrets (using reversible Type 7 algorithms for plain strings and MD5 Type 5 for sensitive fields).

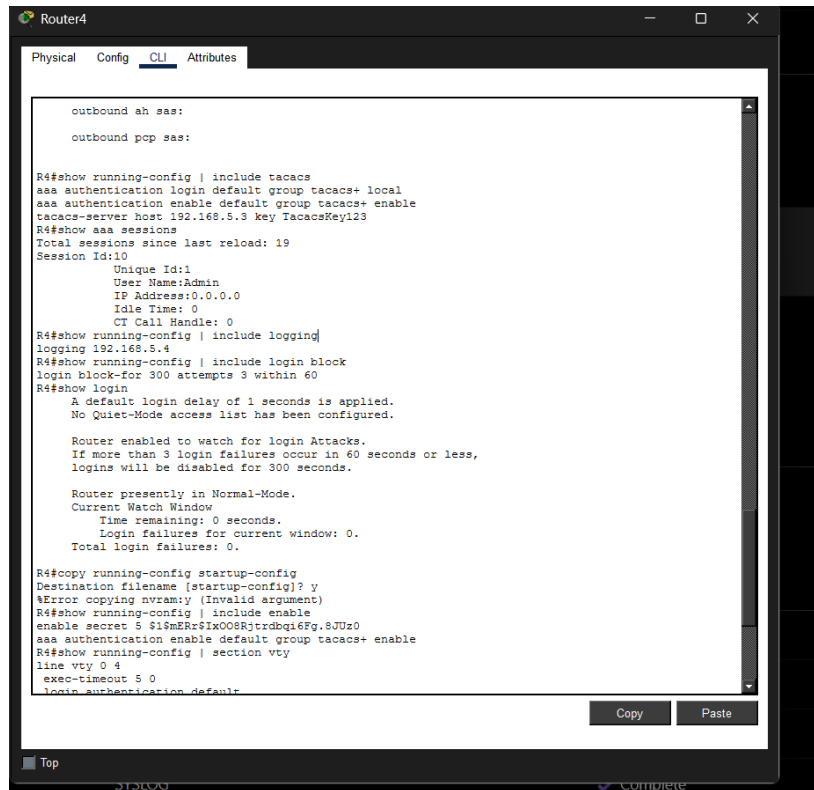


Figure 44: Universal Hashing and Service Password Encryption Verification

CLI: Universal Password Encryption Verification on R4

```

R4#show running-config | include enable
enable secret 5 $1$mERr$Ix008Rjtrdbqi6Fg.8JUz0
  
```

9.10 Configuration 10: Subnet Allocation (192.168.5.0/24)

All internal hosts, servers, and management interfaces in the new branch operational network were allocated and configured under the designated classless IP subnet range '192.168.5.0/24'.

```

Router4
Physical Config CLI Attributes

R4#show running-config | include login block
login block-for 300 attempts 3 within 60
R4#show login
  A default login delay of 1 seconds is applied.
  No Quiet-Mode access list has been configured.

  Router enabled to watch for login Attacks.
  If more than 3 login failures occur in 60 seconds or less,
  logins will be disabled for 300 seconds.

  Router presently in Normal-Mode.
  Current Watch Window
  Time remaining: 0 seconds.
  Login failures for current window: 0.
  Total login failures: 0.

R4#copy running-config startup-config
Destination filename [startup-config]? y
^Error copying nvram: y (Invalid argument)
R4#show running-config | include enable
enable secret 5 $1$mERr$1x008Rjtrdbq16Fg.$JUz0
aaa authentication enable default group tacacs+ enable
R4#show running-config | section vty
line vty 0 4
  exec-timeout 5 0
  login authentication default
  transport input ssh
R4#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R4#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  192.168.5.1    YES NVRAM  up      up
GigabitEthernet0/1  unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2  unassigned      YES NVRAM  administratively down down
Serial0/0/0        172.16.3.254   YES NVRAM  up      up
Serial0/0/1        unassigned      YES NVRAM  down    down
Serial0/1/0        unassigned      YES NVRAM  down    down
Serial0/1/1        unassigned      YES NVRAM  down    down
Serial0/2/0        unassigned      YES NVRAM  down    down
Serial0/2/1        unassigned      YES NVRAM  down    down
Serial0/3/0        unassigned      YES NVRAM  down    down
Serial0/3/1        unassigned      YES NVRAM  down    down
Vlan1            unassigned      YES unset  administratively down down
R4#
  
```

Figure 45: Subnet Allocation Interface Status Verification

CLI: Subnet Interface Brief Verification on R4

```

R4#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  192.168.5.1    YES NVRAM  up      up
GigabitEthernet0/1  unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2  unassigned      YES NVRAM  administratively down down
Serial0/0/0        172.16.3.254   YES NVRAM  up      up
Serial0/0/1        unassigned      YES NVRAM  down    down
Serial0/1/0        unassigned      YES NVRAM  down    down
Serial0/1/1        unassigned      YES NVRAM  down    down
Serial0/2/0        unassigned      YES NVRAM  down    down
Serial0/2/1        unassigned      YES NVRAM  down    down
  
```

```
Serial0/3/0          unassigned      YES  NVRAM  down
                    down
Serial0/3/1          unassigned      YES  NVRAM  down
                    down
Vlan1                unassigned      YES  unset
administratively down down
```

10 Conclusion

Milestone 2 successfully demonstrates the secure expansion of the enterprise network. By implementing a multi-layered security approach—ranging from physical modularity and port-level security to centralized AAA and encrypted inter-branch VPNs—we have established a robust foundation for the new branch operations. All services are fully convergent, monitored via Syslog, and protected against unauthorized access.

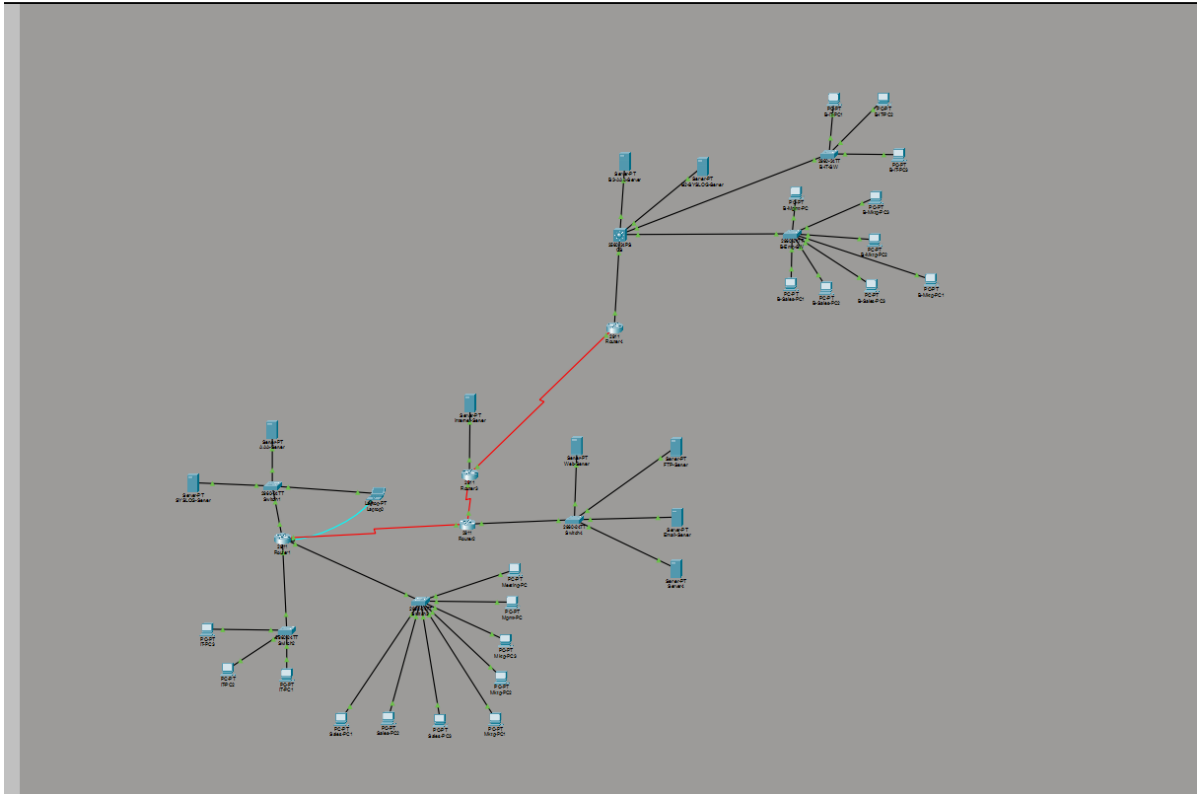


Figure 46: Final Converged Network Infrastructure (Branches 1 2).